

Exercise 7: Electric Fields in Matter 2

Question 1:

A dielectric slab is inserted between the plates of a charged parallel-plate capacitor connected to a battery. The battery maintains a fixed free surface charge density on the plates.

- (a) Explain clearly the difference between free charge and bound charge in this scenario.
(b) Starting from Gauss's law in matter,

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho,$$

explain why the electric displacement field $\mathbf{D}$ depends only on free charge and not explicitly on bound charge.

- (c) When a dielectric is present, explain why applying Gauss's law to $\mathbf{E}$ is often difficult in highly symmetric systems (spherical, cylindrical, planar symmetry). And explain why using

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f\text{enc}}$$

is often simpler?

Question 2:

A thick spherical shell of linear dielectric material occupies the region

$$a < r < b,$$

with permittivity ϵ . A total free charge $+Q_f$ is placed at the center (a point charge at $r = 0$). Assume electrostatic equilibrium and spherical symmetry.

- (a) Use Gauss's law for $\mathbf{D}$,

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f\text{enc}},$$

to find $\mathbf{D}(r)$ in the three regions: (i) $0 < r < a$, (ii) $a < r < b$, (iii) $r > b$.

- (b) Use $\mathbf{D} = \epsilon_0 \mathbf{E}$ in vacuum and $\mathbf{D} = \epsilon \mathbf{E}$ in the dielectric to find $\mathbf{E}(r)$ everywhere.

Question 3:

A linear, isotropic dielectric material is placed in a uniform external electric field $\mathbf{E}_0$. The material is characterized by an electric susceptibility χ_e .

(a) Explain clearly the physical meaning of the electric susceptibility χ_e . Describe what happens microscopically inside the material when an electric field is applied. Explain what it means physically if χ_e is large.

(b) For a linear dielectric, the polarization is given by

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}.$$

Using the definition

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P},$$

derive an expression for:

1. The permittivity ϵ of the material.
2. The dielectric constant (relative permittivity) ϵ_r .

Clearly explain the difference between absolute permittivity ϵ and relative permittivity ϵ_r .

(c) Suppose a material has a very large dielectric constant $\epsilon_r \gg 1$.

Explain qualitatively:

1. What happens to the polarization $\mathbf{P}$ inside the material?
2. How does the electric field inside the material compare to the electric field in vacuum for the same free charge distribution?
3. Why are materials with large dielectric constants useful in capacitors?

Question 4:

An infinitely long straight wire carries a uniform free charge per unit length λ_f . The wire is surrounded by a solid linear, isotropic dielectric cylinder of radius R . The dielectric has electric susceptibility χ_e . Outside the cylinder is vacuum. Assume electrostatic equilibrium and cylindrical symmetry.

(a) Using Gauss's law for $\mathbf{D}$,

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f\text{enc}},$$

determine the displacement field $\mathbf{D}(r)$ for $r < R$ and $r > R$. Express your answer in terms of λ_f .

(b) Using $\mathbf{D} = \epsilon \mathbf{E}$ and $\epsilon = \epsilon_0(1 + \chi_e)$, determine the electric field: inside the dielectric ($r < R$) and outside the dielectric ($r > R$). Show explicitly how the result depends on $\epsilon_r = 1 + \chi_e$. Compare the field inside the dielectric to the field that would exist if the entire space were vacuum.

(c) Using

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E},$$

determine the polarization inside the dielectric. Show explicitly how $\mathbf{P}$ depends on χ_e .

Explain physically what happens when: $\chi_e \rightarrow 0$ and $\chi_e \gg 1$.

(d) Calculate the bound volume charge density ρ_b , and the bound surface charge density σ_b at $r = R$. Explain how the magnitude of the bound surface charge depends on χ_e .

(e) Suppose $\epsilon_r \gg 1$. Explain qualitatively:

1. How the electric field inside the dielectric compares to vacuum.
2. How the polarization changes.
3. Why materials with large ϵ_r are effective at screening electric fields.

Question 5:

A material's response to an electric field is often described by its relative permittivity ϵ_r . In optics, materials are described by their refractive index n . Assume all materials here are non-magnetic, so $\mu_r \approx 1$.

(a) Give approximate values (order of magnitude is fine) of the relative permittivity ϵ_r for natural/common materials we see around us, for example: air, water, glass, plastics, wood (or paper), and a ceramic (if you know one). State the typical range of ϵ_r you would expect for "standard" dielectric materials.

Hint: Here is a website that gives n and ϵ_r (<https://refractiveindex.info/>), ϵ_r is stated as ϵ_1 vs wavelength) of various materials, explore freely! Consider the optical part of the electromagnetic spectrum. Click the blue [i] to the right of the page to find more information of the values that are defined (includes Wikipedia links).

(b) For electromagnetic waves in a linear, homogeneous, non-magnetic medium, refractive index n is described as $n \approx \sqrt{\epsilon_r}$. Use this to check whether typical optical values like glass $n \approx 1.5$, water ≈ 1.33 are consistent with reasonable values of ϵ_r .

Hint: Have a look at <https://refractiveindex.info/>

(c) Describe what is characteristic about the relative permittivity of metals compared to standard dielectrics.

- Is it possible for ϵ_r to be negative?
- What physical mechanism inside metals leads to this behavior?

- Briefly connect this to plasmonics (what kind of phenomenon becomes possible when $\epsilon_r < 0$ at certain wavelength?).

*Hint: have a look on first page of the **ref:** [Literature](#) and Figure 1a.*